GATE — Linear Algebra

Transcribed Handwritten Notes

This document is a clean, searchable transcription of a set of 19 handwritten pages of Linear Algebra notes (matrix and determinant theory, matrix types, rank, elementary operations and echelon form), together with the worked examples and practice questions written in the margins. All mathematics has been typed out so the notes can be read, searched and studied by a person or by an AI assistant. Points where the handwriting was difficult to read are listed at the very end.

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1 Matrices and Determinants: Basics

- A **matrix** is a type of arrangement of numbers in a rectangular form.
- A **determinant** is a *value* (a number), not an arrangement.
- General matrix notation:

$$A = [a_{ij}]_{m\times n} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}_{m\times n}$$

Example of a 4×4 matrix:

$$\begin{bmatrix} 2 & -1 & 4 & 3 \\ 0 & 4 & 1 & 0 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 3 \end{bmatrix}_{4\times 4}$$

- Positional naming based on the indices i (row) and j (column):
  - if $i = j$: **diagonal** element,
  - if $i > j$: **lower-diagonal** element,
  - if $i < j$: **upper-diagonal** element.

Note 1. The value of a determinant can be real or complex — but this is only possible for a *square* matrix.
Note 2. $\mathrm{tr}(A) = \sum_i a_{ii}$ (sum of the diagonal elements) and is called the **trace** of A.

2 Evaluating a Determinant

2.1 Sign pattern (cofactor expansion)

To find the value of a determinant we follow a fixed sign pattern. The sign of the element a_{ij} is

$$\mathrm{sign}(a_{ij}) = (-1)^{i+j}.$$

The sign matrices for orders $2, 3, 4$ are:

$$\begin{vmatrix} + & - \\ - & + \end{vmatrix}, \quad \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}, \quad \begin{vmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{vmatrix}$$

Signs of individual elements:

$$\mathrm{sign}(a_{21}) = (-1)^{2+1} = -1, \quad \mathrm{sign}(a_{23}) = (-1)^{2+3} = -1,$$

$$\mathrm{sign}(a_{43}) = (-1)^{4+3} = -1, \quad \mathrm{sign}(a_{44}) = (-1)^{4+4} = +1.$$

2.2 Worked examples

$$|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$\begin{vmatrix} 2 & -3 \\ 4 & -1 \end{vmatrix} = (2)(-1) - (-3)(4) = -2 + 12 = 10$$

$$\begin{vmatrix} 2 & 0 \\ 3 & -i \end{vmatrix} = (2)(-i) - (3)(0) = -2i$$

$$\begin{vmatrix} 2 & -3 & 1 \\ 1 & 0 & 2 \\ -1 & 2 & 3 \end{vmatrix} = -1 \begin{vmatrix} -3 & 1 \\ 2 & 3 \end{vmatrix} + 0 \begin{vmatrix} 2 & 1 \\ -1 & 3 \end{vmatrix} - 2 \begin{vmatrix} 2 & -3 \\ -1 & 2 \end{vmatrix}$$
$$= -1(-9-2) + 0 - 2(4-3) = 11 + 0 - 2 = 9.$$

2.3 Sarrus' rule (for a 3×3 determinant)

Sarrus' rule: repeat the first two columns to the right of the determinant and take the sum of the three "downward" products minus the sum of the three "upward" products.

$$\begin{vmatrix} 2 & -3 & 1 \\ 1 & 0 & 2 \\ -1 & 2 & 3 \end{vmatrix} = (0 + 6 + 2) - (0 + 8 - 9) = 8 + 1 = 9.$$

$$|A| = \begin{vmatrix} 1 & 2 & 4 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = (45 + 84 + 128) - (140 + 48 + 72) = 257 - 260 = -3.$$

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = (45 + 84 + 96) - (105 + 48 + 72) = 225 - 225 = 0.$$

3 Singular, Non-Singular and Invertible Matrices

- **Singular matrix:** if $|A| = 0$, then A is called *singular*.
- **Non-singular matrix:** if $|A| \neq 0$, then A is called *non-singular*.
- **Invertible matrix:** if A^{-1} exists, then A is called *invertible*, and

$$A^{-1} = \frac{\operatorname{adj} A}{|A|} = \frac{(\operatorname{cof} A)^{\mathsf{T}}}{|A|}.$$

The necessary condition for a matrix to be invertible is $|A| \neq 0$, i.e. the matrix must be *non-singular*; otherwise A^{-1} does not exist.

4 Special Determinants

4.1 A 4×4 determinant

$$|A| = \begin{vmatrix} 0 & 1 & 0 & 1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ -1 & 0 & -1 & 0 \end{vmatrix} = +2 + 2 = 4 = (2)^2.$$

$\because$ A is a **skew-symmetric matrix of even order**, so $|A|$ is a **perfect square**.

4.2 Triangular, diagonal and scalar matrices

To find the determinant of an upper-triangular matrix (U.T.M.), a lower-triangular matrix (L.T.M.), a diagonal matrix, a scalar matrix or an identity matrix, we can simply **multiply the diagonal elements**.

(i) Upper-triangular matrix

$$\begin{bmatrix} 2 & -1 & 4 & 3 \\ 0 & 4 & 1 & 0 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 3 \end{bmatrix} \quad \text{trace} = 8, \quad \det = 2 \cdot 4 \cdot (-1) \cdot 3 = -24, \quad A^{-1} \text{ exists.}$$

(ii) A singular example

$$\begin{bmatrix} -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 2 & -1 & 3 & 0 \\ 0 & 1 & 2 & 1 \end{bmatrix} \quad \det = 0, \quad \text{trace} = 2, \quad A^{-1} \text{ does not exist.}$$

(iii) Identity matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \det = 1, \quad \text{trace} = 4, \quad A^{-1} \text{ exists.}$$

(iv) Scalar matrix $7I_4$

$$\begin{bmatrix} 7 & 0 & 0 & 0 \\ 0 & 7 & 0 & 0 \\ 0 & 0 & 7 & 0 \\ 0 & 0 & 0 & 7 \end{bmatrix} \quad \det = 7^4, \quad \text{trace} = 28, \quad A^{-1} \text{ exists.}$$

5 Worked Example: Roots of Unity and Trace

Question. Let $\alpha = e^{2\pi i/5}$ and let the matrix

$$M = \begin{bmatrix} 1 & \alpha & \alpha^2 & \alpha^3 & \alpha^4 \\ 0 & \alpha & \alpha^2 & \alpha^3 & \alpha^4 \\ 0 & 0 & \alpha^2 & \alpha^3 & \alpha^4 \\ 0 & 0 & 0 & \alpha^3 & \alpha^4 \\ 0 & 0 & 0 & 0 & \alpha^4 \end{bmatrix}_{5\times 5}$$

Then the trace of the matrix $\left(I + M + M^2\right)$ is _____.

Solution.

$$\text{tr}\left(I + M + M^2\right) = \text{tr}(I) + \text{tr}(M) + \text{tr}(M^2).$$

$$\text{tr}(I) = 5, \qquad \text{tr}(M) = 1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 = 0,$$

$$\operatorname{tr}(M^2) = 1 + \alpha^2 + \alpha^4 + \alpha^6 + \alpha^8 = 1 + \alpha^2 + \alpha^4 + \alpha + \alpha^3 = 0$$

(since $\alpha^5 = 1$, so $\alpha^6 = \alpha$ and $\alpha^8 = \alpha^3$; and $1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 = 0$).

$$\therefore\ \operatorname{tr}\big(I + M + M^2\big) = 5 + 0 + 0 = \boxed{5}.$$

6 Inverse of a Matrix

6.1 Shortcut method for a 2×2 matrix

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$,

$$\operatorname{adj} A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, \qquad A^{-1} = \frac{1}{|A|}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Steps: (1) exchange the two diagonal entries, (2) put a minus sign before the two remaining entries.

$$A = \begin{bmatrix} 2 & 0 \\ 1 & -3 \end{bmatrix}, \qquad |A| = -6,$$

$$A^{-1} = \frac{1}{-6}\begin{bmatrix} -3 & 0 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ \frac{1}{6} & -\frac{1}{3} \end{bmatrix}.$$

$$M = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{bmatrix} \Rightarrow\ M^{-1} = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}.$$

6.2 Shortcut method for a 3×3 matrix

For a 3×3 matrix, $A^{-1} = \dfrac{\operatorname{adj} A}{|A|} = \dfrac{(\operatorname{cof} A)^{\mathsf{T}}}{|A|}$, where $\operatorname{cof} A$ is the matrix of cofactors and $\operatorname{adj} A = (\operatorname{cof} A)^{\mathsf{T}}$.

$$A = \begin{bmatrix} 2 & -3 & 1 \\ 1 & 0 & 2 \\ -1 & 2 & 3 \end{bmatrix}, \qquad |A| = 9 \neq 0\ \Rightarrow\ A^{-1} \text{ exists.}$$

$$\operatorname{cof} A = \begin{bmatrix} -4 & -5 & 2 \\ 11 & 7 & -1 \\ -6 & -3 & 3 \end{bmatrix}, \qquad A^{-1} = \frac{(\operatorname{cof} A)^{\mathsf{T}}}{|A|} = \frac{1}{9}\begin{bmatrix} -4 & 11 & -6 \\ -5 & 7 & -3 \\ 2 & -1 & 3 \end{bmatrix}.$$

7 Transpose, Conjugate and Transjugate

For $A = [a_{ij}]_{m\times n}$:

- **Transpose:** $A^{\mathsf{T}} = [a_{ji}]_{n\times m}$
- **Conjugate:** $\overline{A} = [\,\overline{a_{ij}}\,]_{m\times n}$
- **Transjugate:** $A^{\theta} = (\overline{A})^{\mathsf{T}} = \overline{(A^{\mathsf{T}})} = [\,\overline{a_{ji}}\,]_{n\times m}$

$$A = \begin{bmatrix} 2 & 4-i & 7i \\ 8+2i & 0 & -5 \end{bmatrix}$$

$$\overline{A} = \begin{bmatrix} 2 & 4+i & -7i \\ 8-2i & 0 & -5 \end{bmatrix}, \qquad A^{\mathsf{T}} = \begin{bmatrix} 2 & 8+2i \\ 4-i & 0 \\ 7i & -5 \end{bmatrix},$$

$$A^{\theta} = \begin{bmatrix} 2 & 8-2i \\ 4+i & 0 \\ -7i & -5 \end{bmatrix}.$$

If $z = x + iy$ then $\overline{z} = x - iy$: to find the conjugate, replace i by $(-i)$.

8 Reversal Laws and Rules of Matrix Multiplication

Reversal law:

1. $(ABC)^{\mathsf{T}} = C^{\mathsf{T}}B^{\mathsf{T}}A^{\mathsf{T}}, \quad (ABC)^{\theta} = C^{\theta}B^{\theta}A^{\theta}, \quad (ABC)^{-1} = C^{-1}B^{-1}A^{-1}$
2. $(AB)^{-1} = B^{-1}A^{-1}$

- For AB (with $A_{m\times n}$ and $B_{n\times p}$) to be defined, the necessary condition is: *number of columns of the first = number of rows of the second.*
- For $A + B$ (and $A - B$) to be defined, the matrices must be of the *same order*.

If $A_{2\times 3}$, $B_{2\times 3}$, $C_{3\times 3}$, then $A + B$ is defined, but $A + C$ and $B + C$ are *not* defined.

- Condition for A/B (matrix division): **not defined** (N.D.).
- Matrix multiplication is *not commutative* in general: $AB \neq BA$ (in general).

9 Commuting Matrices and $AB = BA = I$

Question. The matrices $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ and $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ commute under multiplication.

$$AB = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} = \begin{bmatrix} a\cos\theta & -b\sin\theta \\ a\sin\theta & b\cos\theta \end{bmatrix}$$

$$BA = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} a\cos\theta & -a\sin\theta \\ b\sin\theta & b\cos\theta \end{bmatrix}$$

For $AB = BA$ we need $-b\sin\theta = -a\sin\theta$ and $a\sin\theta = b\sin\theta$:

$$(a-b)\sin\theta = 0 \implies a = b \quad \text{or} \quad \sin\theta = 0,$$

$$\therefore\ a = b \quad \text{or} \quad \theta = n\pi, \quad n \in \mathbb{Z}.$$

If $AB = C$, where A and B are non-singular matrices, then $B = ?$

$$A^{-1}(AB) = A^{-1}C \ \Rightarrow\ IB = A^{-1}C \ \Rightarrow\ B = A^{-1}C.$$

If $AB = BA = I$, then $A^{-1} = ?$, $B^{-1} = ?$

$$A^{-1} = B, \qquad B^{-1} = A.$$

If $EF = G$, then $F = ?$ where

$$E = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}, \qquad G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$E^{-1}(EF) = E^{-1}G \;\Rightarrow\; F = E^{-1}G = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}.$$

10 Types of Matrices

- **Symmetric matrix:** if $A^{\mathsf{T}} = A$.
- **Skew-symmetric matrix:** if $A^{\mathsf{T}} = -A$.
- **Hermitian matrix:** if $A^{\theta} = A$.
- **Skew-Hermitian matrix:** if $A^{\theta} = -A$.
- **Orthogonal matrix:** if $AA^{\mathsf{T}} = I$, equivalently $A^{-1} = A^{\mathsf{T}}$.
- **Unitary matrix:** if $AA^{\theta} = I$, equivalently $A^{-1} = A^{\theta}$.
- **Nilpotent matrix:** if $A^k = 0$ for the least positive integer k; then A is a nilpotent matrix of order k.
- **Idempotent matrix:** if $A^2 = A$. (Since $I^2 = I$, the identity matrix I is idempotent.)
- **Involutory matrix:** if $A^2 = I$.

10.1 Worked examples on matrix types

Orthogonal matrix. For

$$Q = \frac{1}{7}\begin{bmatrix} 3 & 2 & 6 \\ -6 & 3 & 2 \\ 2 & 6 & -3 \end{bmatrix}, \qquad Q^{-1} = Q^{\mathsf{T}} = \frac{1}{7}\begin{bmatrix} 3 & -6 & 2 \\ 2 & 3 & 6 \\ 6 & 2 & -3 \end{bmatrix}.$$

(Since Q is orthogonal, $Q^{-1} = Q^{\mathsf{T}}$.)

Symmetric: $A = \begin{bmatrix} 2 & -3 & 4 \\ -3 & 0 & 5 \\ 4 & 5 & 1 \end{bmatrix}$ is symmetric.

Skew-symmetric: $A = \begin{bmatrix} 0 & -3 & 5 \\ 3 & 0 & 1 \\ -5 & -1 & 0 \end{bmatrix}$ is skew-symmetric.

In a skew-symmetric matrix the diagonal elements are all 0, and its determinant is also 0 (for odd order).

Hermitian:

$$A = \begin{bmatrix} 2 & 5-i & -7 \\ 5+i & -2 & 3i \\ -7 & -3i & 5 \end{bmatrix}, \quad \overline{A} = \begin{bmatrix} 2 & 5+i & -7 \\ 5-i & -2 & -3i \\ -7 & 3i & 5 \end{bmatrix},$$

$$(\overline{A})^{\mathsf{T}} = \begin{bmatrix} 2 & 5+i & -7 \\ 5-i & -2 & 3i \\ -7 & -3i & 5 \end{bmatrix} = A \;\Rightarrow\; A^{\theta} = A \;\Rightarrow\; A \text{ is Hermitian.}$$

Skew-Hermitian:

$$A = \begin{bmatrix} 2i & 5+i & -7 \\ -5+i & -2i & 3i \\ 7 & 3i & 5i \end{bmatrix}, \quad \overline{A} = \begin{bmatrix} -2i & 5-i & -7 \\ -5-i & 2i & -3i \\ 7 & -3i & -5i \end{bmatrix},$$

$$(\overline{A})^{\mathsf{T}} = \begin{bmatrix} -2i & -5-i & 7 \\ 5-i & 2i & -3i \\ -7 & -3i & -5i \end{bmatrix} = -\begin{bmatrix} 2i & 5+i & -7 \\ -5+i & -2i & 3i \\ 7 & 3i & 5i \end{bmatrix} = -A,$$

$$(\overline{A})^{\mathsf{T}} = -A \;\Rightarrow\; A \text{ is skew-Hermitian.}$$

Determinant of a skew-symmetric matrix. For $A = \begin{bmatrix} 0 & -3 & 5 \\ 3 & 0 & 1 \\ -5 & -1 & 0 \end{bmatrix}$,

$$|A| = 0\begin{vmatrix} 0 & 1 \\ -1 & 0 \end{vmatrix} + 3\begin{vmatrix} -3 & 5 \\ -1 & 0 \end{vmatrix} + 5\begin{vmatrix} -3 & 0 \\ -5 & -1 \end{vmatrix} = 0 + 3(0+5) + 5(3-0) = 15 - 15 = 0.$$

Orthogonal (rotation) matrix. $M = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{bmatrix}$ is orthogonal:

$$M^{\mathsf{T}} = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}, \qquad MM^{\mathsf{T}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

Unitary matrix.

$$A = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix} \;\Rightarrow\; A \text{ is unitary.}$$

$$A^{\mathsf{T}} = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix}, \qquad A^{\theta} = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \overline{\omega} & \overline{\omega}^2 \\ 1 & \overline{\omega}^2 & \overline{\omega} \end{bmatrix}.$$

Here $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$, $\omega^2 = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$, with

$$\overline{\omega} = \omega^2, \quad \overline{\omega}^2 = \omega, \quad \omega^3 = 1, \quad 1 + \omega + \omega^2 = 0.$$

$$\therefore\; AA^{\theta} = \frac{1}{3}\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I.$$

Nilpotent matrix. $A = \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix}$:

$$A^2 = \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0,$$

i.e. A is a nilpotent matrix of order 2.

If A is a 2×2 skew-symmetric matrix then:

1. $|A| = 0$ if the order is odd; $|A|$ is a perfect square if the order is even.
2. $\sum a_{ij} = 0$ whether the order is odd or even.

$A = \begin{bmatrix} 0 & -2 & 1 \\ 2 & 0 & 4 \\ -1 & -4 & 0 \end{bmatrix}_{3\times 3}$ with $\sum a_{ij} = 0$:

$$|A| = 0 \begin{vmatrix} 0 & 4 \\ -4 & 0 \end{vmatrix} - (-2) \begin{vmatrix} 2 & 4 \\ -1 & 0 \end{vmatrix} + 1 \begin{vmatrix} 2 & 0 \\ -1 & -4 \end{vmatrix} = 0 + 8 - 8 = 0.$$

The maximum number of terms in the general expansion of $|A|_{n\times n}$ is $n!$.

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \;\Rightarrow\; 2 \text{ terms} = 2!$$

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a(ei - hf) - b(di - gf) + c(dh - eg) \;\Rightarrow\; 6 \text{ terms} = 3!,$$

and so on.

The maximum number of *different* elements needed to construct a symmetric matrix of order n is $\dfrac{n(n+1)}{2}$:

$$A_{2\times 2} = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \Rightarrow 3, \quad A_{3\times 3} = \begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix} \Rightarrow 6,$$

$$A_{4\times 4} = \begin{bmatrix} a & b & c & d \\ b & e & f & g \\ c & f & h & i \\ d & g & i & j \end{bmatrix} \Rightarrow 10,$$

and so on.

Question. If A is a Hermitian matrix, then iA is:
(a) Symmetric (b) Skew-symmetric (c) Hermitian (d) Skew-Hermitian

Solution. Given $A^\theta = A \quad \cdots(1)$.

$$(iA)^\theta = \bar{i}\, A^\theta = (-i)A = -iA = -(iA).$$

Note: if the transjugate of a matrix comes out with a negative sign, then it is a **skew-Hermitian** matrix. $\therefore$ **Answer: (d) Skew-Hermitian.**

11 Rank

The **rank** of a matrix is the order of the largest non-singular sub-matrix that can exist in the given matrix.

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & -1 \end{bmatrix} \Rightarrow |A| \neq 0, \quad \text{so} \quad \rho(A) = 3.$$

A matrix obtained by deleting some rows, some columns, or both, is called a **sub-matrix**.

$$A = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 4 & -6 \\ -1 & 2 & -3 \end{bmatrix}_{3\times 3}$$

$\because |A| = 0$, so $\rho(A) \neq 3$. All the 2×2 sub-matrices are also singular, hence $\rho(A) = 1$.

$$A = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 4 & -6 \\ -1 & 2 & 7 \end{bmatrix}_{3\times 3}$$

$\because |A| = 0 \Rightarrow \rho(A) \neq 3$. There is a 2×2 sub-matrix $A_1 = \begin{bmatrix} 4 & -6 \\ 2 & 7 \end{bmatrix}$ (obtained by deleting R_1 and C_1) which is non-singular, $\therefore |A_1| \neq 0$, so $\rho(A) =$ order of $A_1 = 2$.

$$A = \begin{bmatrix} 2 & -1 & 3 & 4 & 1 \\ -2 & 1 & -5 & -4 & 1 \\ 0 & 0 & 7 & 0 & 0 \end{bmatrix}_{3\times 5}$$

$\rho(A) \neq 3$, $\because$ all the 3×3 sub-matrices are singular. Most of the 2×2 sub-matrices are also singular, but there is a sub-matrix $A_1 = \begin{bmatrix} 1 & -3 \\ 0 & 7 \end{bmatrix}_{2\times 2}$ (obtained by deleting C_1, C_4, C_5 and R_1) with $|A_1| = 7 \neq 0$ (i.e. A_1 is non-singular), so $\rho(A) = 2$.

Note: $\rho(A) \neq 5$ or 4, $\because$ it is not possible to obtain a sub-matrix of order 5×5 or 4×4.

Question. The rank of the matrix

$$\begin{bmatrix} 0 & i & -i \\ -i & 0 & i \\ i & -i & 0 \end{bmatrix}$$

is: (a) 1 (b) 2 (c) 3 (d) 4

$\because$ in a skew-symmetric matrix of odd order the diagonal elements are 0 and the determinant is also 0, i.e. $\rho(A) \neq 3$. But the 2×2 sub-matrix

$$A_1 = \begin{bmatrix} i & -i \\ 0 & i \end{bmatrix} \Rightarrow |A_1| = i^2 - 0 = -1 \neq 0,$$

$\therefore A_1$ is non-singular $\Rightarrow \rho(A) = 2$. **Answer: (b).**

11.1 Properties of Rank

1. $\rho(A_{m\times n}) \le \min\{m, n\}$ — the rank of $A_{m\times n}$ is less than or equal to the minimum of its number of rows and columns. E.g. $\rho(A_{3\times 5}) \le \min\{3, 5\} \Rightarrow \rho(A_{3\times 5}) \le 3$.
2. $1 \le \rho(A_{m\times n}) \le \min\{m, n\}$.
3. If $A_{n\times n}$ is non-singular then $\rho(A)$ equals the order of the matrix; a null matrix has rank 0 (by definition).
4. $\rho(A) = \rho(A^{-1}) = \rho(A^{\mathsf{T}}) = \rho(A^{\theta}) = \rho(AA^{\mathsf{T}}) = \rho(AA^{\theta})$.
5. The rank of a product cannot be bigger than the ranks of the individual matrices:

$$\rho(AB) \le \min\{\rho(A), \rho(B)\}.$$

Question. The rank of the 4×4 skew-symmetric matrix

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ -1 & 0 & -1 & 0 \end{bmatrix}$$

is _____.

(Left blank in the original notes, to be solved later.)

Solution. Since A is a skew-symmetric matrix of even order, $|A|$ is a perfect square:

$$|A| = 4 = (2)^2 \neq 0 \;\Rightarrow\; A \text{ is non-singular} \;\Rightarrow\; \rho(A) = 4.$$

Answer: 4.

Question. If A and B are matrices of the same order, then:
(a) $\rho(A + B) \le \rho(A) + \rho(B)$
(b) $\rho(A + B) \ge \rho(A) + \rho(B)$
(c) $\rho(A + B) = \rho(A) + \rho(B)$
(d) None of the above

Answer: (a) $\rho(A + B) \le \rho(A) + \rho(B)$.

Question. Let $P = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$ and $Q = \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix}$ be two matrices. Then the rank of $P + Q$ is _____.

$$P + Q = \begin{bmatrix} 0 & -1 & -2 \\ 8 & 9 & 10 \\ 8 & 8 & 8 \end{bmatrix}, \qquad \rho(P + Q) = 2.$$

11.2 Additional Practice Questions

Question. The rank of a 3×3 matrix $C = AB$, found by multiplying a non-zero column matrix A of size 3×1 and a non-zero row matrix B of size 1×3, is:
(a) 0 (b) 1 (c) 2 (d) 3

Solution. $A_{3\times 1}$ is non-zero $\Rightarrow \rho(A) = 1$, and $B_{1\times 3}$ is non-zero $\Rightarrow \rho(B) = 1$.

$$C = AB = (AB)_{3\times 3}, \qquad \rho(C) = \rho(AB) \le \min\{\rho(A), \rho(B)\} = 1.$$

Also $C \neq 0$ (the product of a non-zero column and a non-zero row is non-zero), so $\rho(C) \geq 1$. Hence $\rho(C) = 1$. **Answer: (b).**

Question. The rank of the matrix

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

is: (a) 1 (b) 2 (c) 3 (d) 4

Solution. Apply $R_4 \to R_4 - R_3$, $R_3 \to R_3 - R_2$, $R_2 \to R_2 - R_1$:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 4 & 4 & 4 \\ 4 & 4 & 4 & 4 \\ 4 & 4 & 4 & 4 \end{bmatrix}.$$

Then apply $R_3 \to R_3 - R_2$ and $R_4 \to R_4 - R_2$:

$$A_1 = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 4 & 4 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad \rho(A_1) = 2$$

(the 2×2 sub-matrix $\begin{bmatrix} 1 & 2 \\ 4 & 4 \end{bmatrix}$ is non-singular). Since $A \sim A_1$, $\rho(A) = 2$. **Answer: (b).**

Question. The rank of the matrix

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 3 & 6 & 9 & 12 \\ 0 & 5 & 10 & 15 & 20 \end{bmatrix}$$

is: (a) 0 (b) 1 (c) 2 (d) 3

Solution. Apply $R_3 \to R_3 - R_2$:

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 3 & 6 & 9 & 12 \\ 0 & 2 & 4 & 6 & 8 \end{bmatrix}.$$

Then apply $R_2 \to R_2 - 3R_1$ and $R_3 \to R_3 - 2R_1$:

$$A_1 = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \qquad \rho(A_1) = 1.$$

Since $A \sim A_1$, $\rho(A) = 1$. **Answer: (b).**

Question. If for a matrix the rank equals both the number of rows and the number of columns, then the matrix is called:
(a) non-singular (b) singular (c) transpose (d) minor

Solution. If $A_{n\times n}$ has $\rho(A) = n$, i.e. $\rho(A)$ = (number of rows) = (number of columns), then

$$\rho(A) = n \;\Rightarrow\; |A| \neq 0 \;\Rightarrow\; A \text{ is non-singular.}$$

Answer: (a).

12 Elementary Operations and Equivalent Matrices

Elementary row operations:

i. $R_i \leftrightarrow R_j$ (interchange two rows)
ii. $R_i \to kR_i$ (multiply a row by a scalar)
iii. $R_i \to R_i - kR_j$ (add a multiple of one row to another)

Elementary operations do **not** change the rank. So any elementary operation may be performed on any matrix without changing its rank. However, when evaluating a *determinant*, only the third operation may be performed.

Equivalent matrix: the matrix obtained after performing elementary operations is called an equivalent matrix.

Question. Applying an elementary transform to a matrix, its rank _____.
(a) increases (b) decreases (c) does not change (d) None of the above

Answer: (c) — an elementary transform does not change the rank.

13 Echelon Form

A matrix is in **echelon form** if:

1. There is a zero before a non-zero number, and the number of zeros increases in subsequent rows.
2. The zero row, or the row with the most zeros before a non-zero number, is at the bottom.

Note: Any matrix can be converted into echelon form. It is very easy to get the rank from the echelon form — simply count the number of non-zero rows.

Notes on the Transcription

All previously flagged readings have been checked against the original pages and corrected where necessary; no unresolved transcription issues remain.